

Linear Equations in Two Variables

Framing a Linear Equation

Linear equation in one variable

When an equation has only one variable of degree one, then that equation is known as linear equation in one variable.

- Standard form: $ax+b=0$, where a and $b \in R$ & $a \neq 0$
- Examples of linear equation in one variable are :

- $3x-9 = 0$

- $2t = 5$

Linear equation in 2 variables

When an equation has two variables both of degree one, then that equation is known as linear equation in two variables.

Standard form: $ax+by+c=0$, where $a, b, c \in R$ & $a, b \neq 0$

Examples of linear equations in two variables are:

- $7x+y=8$

- $6p-4q+12=0$

Examples of a Linear Equations

Solution of linear equation in 2 variables

A linear equation in two variables has a pair of numbers that can satisfy the equations. This pair of numbers is called as the solution of the linear equation in two variables.

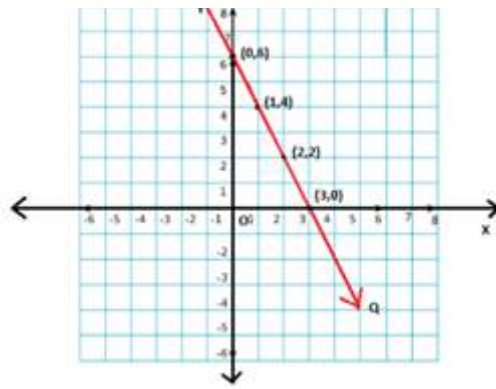
- The solution can be found by assuming the value of one of the variable and then proceed to find the other solution.
- There are infinitely many solutions for a single linear equation in two variables.

Graph of a Linear Equation

Graphical representation of a linear equation in 2 variables

- Any linear equation in the standard form $ax+by+c=0$ has a pair of solutions (x,y) , that can be represented in the coordinate plane.
- When an equation is represented graphically, it is a straight line that may or may not cut the coordinate axes.





Solutions of Linear equation in 2 variables on a graph

- A linear equation $ax+by+c=0$ is represented graphically as a straight line.
- Every point on the line is a solution for the linear equation.
- Every solution of the linear equation is a point on the line.

Lines passing through origin

- Certain linear equations exist such that their solution is $(0,0)$. Such equations when represented graphically pass through the origin.
- The coordinate axes x-axis and y-axis can be represented as $y=0$ and $x=0$ respectively.

Lines parallel to coordinate axes

- Linear equations of the form $y=a$, when represented graphically are lines parallel to the x-axis and a is the y-coordinate of the points in that line.
- Linear equations of the form $x=a$, when represented graphically are lines parallel to the y-axis and a is the x-coordinate of the points in that line.

